📊 5. Analysis Summary: Temperature & Air Quality (AQI)

Atlantic Canada (1995–2025)

🟦 5.1 Overview

This section presents the analysis of **temperature trends and air quality (AQI)** across Atlantic Canada, including Nova Scotia (NS), New Brunswick (NB), Prince Edward Island (PEI), and Newfoundland & Labrador (NL).

The analysis is part of a broader multi-domain dashboard integrating **temperature, precipitation, extreme weather events, greenhouse gas (GHG) emissions, and AQI**, enabling a comprehensive understanding of climate patterns and environmental risks.

🟦 5.2 Temperature Analysis

📌 5.2.1 Long-Term Trend

The analysis identifies a **consistent warming trend of approximately 0.31°C per decade across Atlantic Canada**, based on historical temperature data.

Regional variation:

- PEI shows the highest warming rate (~0.46°C/decade)
- Other provinces show steady but moderate increases

👉 Refer to **Figure : Temperature Trend by Province** (Dashboard – Trend Chart)

This demonstrates that:

Climate change impacts are **region-wide but unevenly distributed**

📌 5.2.2 Variability and Extremes

While average temperature is increasing, the data highlights:

- Greater fluctuations in temperature values year-to-year
- Increased frequency of extreme temperature peaks (above historical norms)

👉 Refer to **Figure: Maximum Temperature Over Time**

This indicates:

Climate change in Atlantic Canada is not only gradual warming but also **increasing variability and unpredictability**

📌 5.2.3 Heatwave Patterns

Heatwave analysis reveals:

- Increasing heatwave frequency across all provinces
- Highest concentrations in:
  - Newfoundland and Labrador (~29/month)
  - Nova Scotia (~27/month)
- PEI exhibits fewer heatwaves despite higher warming

👉 Refer to **Figure: Heatwaves vs Temperature Panel**

This suggests:

Heatwave occurrence is driven by **climate variability and regional conditions**, not just average temperature increase

📌 5.2.4 Interpretation

The temperature analysis confirms:

"Atlantic Canada is warming steadily, but the most critical concern is the rise in variability and frequency of extreme events rather than average temperature changes."

🟩 5.3 Air Quality (AQI) Analysis

📌 5.3.0 Data Scope Note

The AQI analysis differs from other climate datasets in terms of time coverage:

- **Temperature & climate data:** ~1995–2025 (30-year range)

- **AQI data: 2014–2024 (10-year range)**

👉 This difference is due to:

- Availability of standardized AQI datasets
- Consistency of air quality monitoring data (NAPS stations)

"As a result, AQI insights reflect **recent air quality trends and short-term variability**, rather than long-term historical climate patterns."

📌 5.3.1 Overall AQI Trends

The Air Quality Index (AQI) analysis is based on available data from **2014 to 2024**, covering a 10-year period within the overall 30-year climate study (1995–2025).

The average Air Quality Index (AQI) across the region remains classified as **"Good"**.

However:

- A total of **166 unhealthy AQI days** were recorded
- Periodic spikes exceed safe thresholds (>100 and up to ~200+)

👉 Refer to **Figure: AQI KPI & Unhealthy Days Indicators**

This highlights that:

Average AQI values mask **critical short-term risk exposures**

📌 5.3.2 AQI Spikes and Seasonal Behavior

The AQI trends show:

- Seasonal peaks, particularly during warmer months
- Sudden spikes associated with environmental triggers

Major contributing pollutants include:

- Ozone (O_3)
- PM2.5
- PM10

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👉 Refer to **Figure: Worst Pollutant AQI Trend Chart**

📌 5.3.3 Correlation with Temperature

The relationship between temperature and AQI varies regionally:

- Nova Scotia → strong negative correlation (~ -0.43)
- PEI → positive correlation (~ +0.187)
- Other provinces → weak or inconsistent correlations

👉 Refer to **Figure: Temperature–Pollutant Correlation Chart**

This indicates:

AQI is influenced by **multiple environmental and regional factors**, not temperature alone

📌 5.3.4 Drivers of AQI Variability

Key factors influencing AQI spikes include:

- Wildfire activity
- Industrial emissions
- Meteorological conditions (wind patterns, seasonal variations)

👉 These drivers create **episodic but high-impact pollution events**

📌 5.3.5 Interpretation

The AQI analysis demonstrates:

"Air quality risk in Atlantic Canada is event-driven, where short-term spikes significantly impact health and operations despite stable long-term averages."

🟥 5.4 Integrated Temperature–AQI Insights

📌 5.4.1 Cross-Domain Insight

The combined analysis reveals:

- Temperature rise alone does not directly determine AQI levels
- Regions with higher variability experience more extreme environmental events
- AQI spikes are **episodic but disproportionately impactful**

📌 5.4.2 Key Insight

"While Atlantic Canada is warming steadily, the real risk lies in increasing climate variability and extreme environmental events, rather than gradual average change."

👉 Refer to **Figure: Summary Insight Panel (Dashboard Highlight)**

📌 5.4.3 Risk Implications

The interaction between temperature and AQI leads to:

- Higher likelihood of heatwaves
- Increased pollution sensitivity
- Greater unpredictability in environmental conditions

🟨 5.5 Business and Policy Implications

📌 5.5.1 Operational Impact

- Increased energy demand during heatwaves
- Workforce productivity affected during AQI spikes

📌 5.5.2 Public Health Impact

- Higher risk during extreme heat and pollution events
- Need for improved alert and response systems

📌 5.5.3 Infrastructure & Planning

- Increased stress on infrastructure systems
- Need for climate-resilient policy frameworks

📌 5.5.4 Strategic Value

This analysis supports:

- Government agencies (environment, health, emergency planning)
- Energy sector forecasting
- Insurance and risk modeling

🟥 5.6 Conclusion

The analysis confirms that:

- Atlantic Canada is experiencing **consistent warming trends**
- AQI appears stable but is impacted by **significant short-term spikes**
- Environmental risk is increasingly driven by:
  - climate variability
  - extreme events
  - regional differences

It is important to note that AQI insights are based on a shorter time window (2014–2024), and therefore primarily capture **recent trends and variability**, whereas temperature analysis reflects long-term climate behavior.

✅ Final Insight

"Stable averages do not indicate low risk. Increasing variability in temperature and AQI is driving hidden but significant environmental, operational, and health risks across Atlantic Canada."

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